

Internship Project Documentation

C++ Console Applications

This document contains the complete documentation of the internship tasks developed in C++. Each task demonstrates important programming concepts such as arrays, functions, file handling, object-oriented programming, recursion, backtracking, and data management.

Task 1: CGPA Calculator System

Objective: To develop a CGPA Calculator system that calculates student GPA and generates semester reports.

Key Features
Input course names and credit hours
Calculate GPA and CGPA
Letter grade support
Performance evaluation
Pass/Fail status
Best and weakest course detection
Course searching feature

Programming Concepts Used: Arrays, Loops, Conditional Statements, Input Validation, Formatting using iomanip

Task 2: Login and Registration System

Objective: To create a secure user authentication system using file handling.

Key Features
User Registration
User Login
Duplicate username checking
Password validation
Forgot password feature
Delete account functionality
Password update system

Programming Concepts Used: File Handling, Functions, Strings, Validation Techniques

Task 3: Sudoku Solver

Objective: To implement a Sudoku Solver using recursion and backtracking algorithms.

Key Features
9x9 Sudoku grid
Backtracking algorithm
Row, column and box checking
Recursive solving
Custom puzzle input
Formatted board display

Programming Concepts Used: 2D Arrays, Recursion, Backtracking, Nested Loops, Functions

Task 4: Banking Management System

Objective: To develop a banking management system using object-oriented programming concepts.

Key Features
Customer account creation
Deposit and withdrawal
Fund transfer
Transaction history
Balance inquiry
Multiple customer management

Programming Concepts Used: Classes and Objects, Encapsulation, Vectors, Constructors, Composition

Conclusion:
These internship tasks helped in improving practical programming skills in C++. The projects demonstrate problem solving, logical thinking, object-oriented programming, file handling, recursion, and console-based application development.